Jiayu Tang

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RESEARCH INTERESTS

Interested in integrating AI-driven antibody design and structure prediction, experimental validation of antibody-antigen binding, statistical evaluation of model confidence and reliability, and translation of engineered antibodies toward therapeutic use.

EDUCATION

M.S., Food Science and Technology; Minor in Statistics 08/2024 - 08/2026
University of Nebraska-Lincoln, Lincoln, NE, USA GPA 3.95/4.00

B.Sc., Food Science and Technology (Dual Degree) 09/2020 - 06/2024
University of Nebraska-Lincoln, Lincoln, NE, USA GPA 3.52/4.00

B.Eng., Food Science and Engineering (Dual Degree)
Northwest A&F University, China

RESEARCH EXPERIENCE

OPS Research Assistant 10/2026 - Present
Department of Infectious Diseases and Immunology, College of Veterinary Medicine, University of Florida
Supervisor: Dr. Yiquan Wang

- Investigate how human VH-VL framework architecture influences predicted antibody-antigen recognition.
- Develop in-silico antibody-design pipelines built on framework grafting, native-structure controls, and epitope-masked structural evaluation.
- Run large-scale AlphaFold2 antibody-antigen predictions and evaluate model confidence, docking accuracy, and interface contacts.
- Benchmark emerging antibody-design tools, including RFantibody and BindCraft2.
- Use AI coding agents (Claude Code, Codex) to plan, implement, and audit computational workflows.

Graduate Research Assistant 07/2024 - 07/2026
Department of Food Science and Technology, University of Nebraska-Lincoln
Advisor: Dr. Edward Deehan

Clinical trials:

- Coordinated a 12-week, three-phase randomized crossover trial of dry bean market-class effects on gut microbiota and cardiometabolic health markers (IRB Project #: UNL-00024307; ClinicalTrials.gov #: NCT06935435).
- Co-developed the NIH-style study protocol, informed consent form, and recruitment materials; authored 4 standard operating procedures (SOPs) for Deehan Lab.
- Ran 200+ clinic visits and oversaw recruitment, scheduling, documentation, and study supplies.
- Assisted a human biobank study that collect human blood and stool samples with health information (IRB Project #: UNL-00023846).
- Collected and analyzed clinical data in R and Python with multivariate and regression methods.

Systematic review:

- Led a systematic review project to evaluate the effects of dietary pulses market classes on gut microbiome features and health-related outcomes across peer-reviewed acute and long-term intervention clinical trials (PROSPERO Registration #: CRD420261350487).
- Searched databases and screened 1,800+ records with machine-learning tools (ASReview, TERA-Tool).
- Applied a large language model (Gemini 3 Pro) to assist data extraction, with every extracted value verified by two human reviewers.

Undergraduate Research Assistant 04/2022 - 06/2023

College of Food Science and Engineering, Northwest A&F University

Advisor: Dr. Chunxia Xiao

- Co-led an *in vitro* fermentation project testing how three lactic acid bacteria species change the phenolic content, antioxidant activity, and starch digestibility of *proso* millet (Undergraduate Thesis).
- Measured total phenolic content (Folin–Ciocalteu, gallic acid equivalents) and DPPH and ABTS radical-scavenging activity by spectrophotometry (co-author, Foods 2023).
- Synthesized findings from 50+ articles on coarse grain consumption and type 2 diabetes for a review article (co-author, J Sci Food Agric 2023).

PUBLICATIONS

*# co-first author; * corresponding author*

In preparation

1. **Tang J#**, Adinchezor C#, Grimaldo C, Zhang Y, Iwuamadi I, Rasmussen C, Rose D, Majumder K, Deehan E*. Exploring the impact of different dietary pulse market classes on human gut microbiota and health.
2. **Tang J#**, Rasmussen C#, Deehan E*. Effect of distinct dietary pulse market classes on gut microbiome and health markers: a systematic review of intervention trials.
3. Grimaldo C, **Tang J**, Whisenhunt L, Chai W, Deehan E*. Evaluating interrelationships between body composition and immunometabolic dysfunction using bioelectrical impedance analysis: an observational study.

Peer-reviewed

1. Xiao J, Li Y, Niu L, Chen R, **Tang J**, Tong Z, Xiao C*. Effect of adding fermented proso millet bran dietary fiber on micro-structural, physicochemical, and digestive properties of gluten-free proso millet-based dough and cake. Foods. 2023;12(15):2964.
2. Deng X, Niu L, Xiao J, Guo Q, Liang J, **Tang J**, Liu X, Xiao C*. Involvement of intestinal flora and miRNA into the mechanism of coarse grains improving type 2 diabetes: an overview. J Sci Food Agric. 2023;103(9):4257-4267.

AWARDS & SCHOLARSHIPS

Chancellor's Graduate Research Grant Award	2026
Departmental Graduate Research Assistantship	2024 - 2026
Excellence in the Department of Food Science and Technology	2025
Mary and Charles C. Cooper / Emma I. Sharpless Fellowship	2024
National Undergraduate Life Sciences Research Competition, Silver Prize	2023
Undergraduate Merit Scholarship	2020 - 2023

TALKS & PRESENTATIONS

Tang J, "Exploring the impact of different dietary pulse market classes on human gut microbiota and health," *Nebraska Food Innovation Center, UNL (presentation)* 2026

Rasmussen C, **Tang J**, et al., "Effect of different Nebraska dry bean market classes on human health and the gut microbiota: a preliminary analysis," *Nebraska Research Days, UNL (poster)* 2026

Rasmussen C, **Tang J**, et al., "Dietary pulse market class-specific effects on gut microbiota, metabolites, and human health: a systematic review," *Nebraska Research Days, UNL (poster)* 2026

Tang J, et al., "Effect of different Nebraska dry bean market classes on human health and the gut microbiota: a 3-phase randomized crossover clinical trial," *Missouri Valley Branch ASM Annual Meeting (poster)* 2026

Tang J, et al., "Exploring the impact of different dietary pulse market classes on host metabolic health and gut microbiome," *Food Science and Technology Symposium, Nebraska Research Days, and NPOD 16th Annual Spring Retreat, UNL (poster)* 2025

TEACHING EXPERIENCE

Teaching Assistant, FDST 458, FDST 458-153 2026
University of Nebraska-Lincoln
- Led lab sessions and graded coursework.

Guest Lecturer, FDST 498 2025
University of Nebraska-Lincoln
- Gave three lectures on systematic review methods and AI-assisted screening.

Teaching Assistant, FDST 412, FDST 429, NUTR 250 2023 - 2024
University of Nebraska-Lincoln
- Graded assignments and exams.

Teaching Assistant (Internship) 2022 - 2023
Xi'an New Oriental Education & Technology Group
- Coached on IELTS exam preparation.

Teaching Assistant, COMM 209 2022
Northwest A&F University
- Graded assignments and exams.

LEADERSHIP & MENTORSHIP

Leadership

Graduate Representative (Secondary) 2025 - 2026
University of Nebraska-Lincoln
- Advocated for food science graduate students on academic affairs.

Graduate Student Advisory Board Member 2024 - 2025
University of Nebraska-Lincoln
- Founded a peer-mentoring program pairing new graduate students with mentors.

Graduate Representative (Primary) 2024 - 2025
University of Nebraska-Lincoln
- Organized three professional development workshops for graduate students.

Communications Lead 2020 - 2022
Northwest A&F University
- Directed a 13-member team producing weekly newsletters and videos.

Mentorship

Cade Rasmussen, Mentor & Co-supervisor, *University of Nebraska-Lincoln* 2025 - 2026
Luo Lin, Mentor, *University of Nebraska-Lincoln* 2025 - 2026
Yaxian Yang, Mentor, *University of Nebraska-Lincoln* 2025

SERVICE

Volunteer, *Nebraska Food for Health Center* 2025
Academic Advisor, *Northwest A&F University* 2023-2024
Junior Reviewer, *Clinical Nutrition* 2024
Academic Coordinator, *Northwest A&F University* 2021-2023
Volunteer, Office of Undergraduate Studies, *Northwest A&F University* 2022
Volunteer, Rural Poverty Alleviation Program, *Northwest A&F University* 2022

PROFESSIONAL DEVELOPMENT, AFFILIATIONS & MEMBERSHIPS

Certifications

Python for Data Science, AI & Development, *Coursera* 2025
Data Science: R Basics, *HarvardX* 2025
Alberta Food Safety Basics, *University of Alberta Health Service* 2025
GCP for Clinical Trials with Investigational Drugs and Biologics, *CITI Program* 2024
Biomedical Responsible Conduct of Research, *CITI Program* 2024

Conflicts of Interest, *CITI Program* 2023
Biomedical Investigators and Key Personnel, *CITI Program* 2023

Workshops & training

High-Performance Computing for Researchers, *University of Nebraska-Lincoln* 2025
Exploring Your Personal Approach to Leadership & Collaboration, *Clifton Strengths Institute* 2025
Grant Writing Seminar, *University of Nebraska-Lincoln* 2025
Peer Mentorship Program, *Northwest A&F University Psychological Education* 2021

Affiliations & memberships

American Society for Microbiology (ASM) 2024 - 2026
American Society for Nutrition (ASN) 2024 - 2026
International Scientific Association for Probiotics and Prebiotics (ISAPP) 2023 - 2025

SKILLS

Computational antibody modeling: AlphaFold structure prediction; *de novo* antibody-design pipelines (RFantibody, BindCraft2); AI coding agents (Claude Code, Codex) for building and validating workflows
Statistics: experimental design and power analysis; multiple regression and multivariate analysis; interpretation in biological and clinical context
Wet lab: PCR; human blood and stool sample processing; in vitro fermentation; spectrophotometric assays
Clinical research: propose, design, and lead human clinical trials; protocol and SOP development; IRB/GCP procedures; ClinicalTrials.gov registration; participant recruitment and visit coordination; grant proposal writing
Evidence synthesis & communication: systematic review; LLM-assisted data extraction with human validation; grant writing
Languages: Mandarin (native), English (fluent)